

◆ ALPHA TEAM 1.0 ◆
ENGINEERING ADMISSION
PREVIOUS YEAR QUESTIONS
PYQ BOOK



TOGETHER WE LEARN



TOGETHER WE LEAD



TOGETHER WE SUCCEED

Learn • Practice • Succeed

Your Path to the Right Engineering College

Team Alpha

MHTCETSUCCESS85@gmail.com

1. MATHEMATICAL LOGIC

MHT-CET 2021 PYQs

1. The logical expressions $p \wedge (\sim p \vee \sim q) \wedge q$

(A) $p \vee q$

(B) $p \wedge q$

(C) F

(D) T

2. p: It rains today

q : I am going to school

r : I will meet my friend

s: I will go to watch a movie

Then symbolic form of the statement "If it does not rain today or I won't go to school, then I will meet my friend and I will go to watch a movie" is

(A) $\sim (p \vee q) \rightarrow (r \vee s)$

(B) $(p \wedge q) \rightarrow (r \vee s)$

(C) $\sim (p \wedge q) \rightarrow (r \wedge s)$

(D) $(\sim p \wedge q) \rightarrow (r \wedge s)$

3. Negation of $(p \wedge q) \rightarrow (\sim p \vee r)$ is

(A) $p \vee q \vee (\sim r)$

(B) $p \wedge q \wedge r$

(C) $\sim p \wedge q \wedge r$

(D) $p \wedge q \wedge (\sim r)$

4. Negation of the statement $\forall x \in \mathbb{R}, x^2 + 1 = 0$ is

(A) $\exists x \in \mathbb{R}$ such that $x^2 + 1 < 0$

(B) $\exists x \in \mathbb{R}$ such that $x^2 + 1 \leq 0$

(C) $\exists x \in \mathbb{R}$ such that $x^2 + 1 \neq 0$

(D) $\exists x \in \mathbb{R}$ such that $x^2 + 1 = 0$

5. If p, q are true statements and r is false statement, then which of the following is correct?

(A) $(p \vee q) \vee r$ has truth value F

(B) $(p \wedge q) \rightarrow r$ has truth value T

(C) $(p \rightarrow r) \rightarrow q$ has truth value F

(D) $(p \leftrightarrow q) \rightarrow r$ has truth value F

6. The logical statement $(p \rightarrow q) \wedge (q \rightarrow \sim p)$ is equivalent to

(A) $\sim p$

(B) p

(C) q

(D) $\sim q$

7. If $p \rightarrow (\sim p \vee q)$ is false, then the truth values of p and q are, respectively

(A) T, F

(B) F, F

(C) F, T

(D) T, T

8. If statements p and q are true and r and s are false, then truth values of $\sim(p \rightarrow q) \leftrightarrow (r \wedge s)$ and $(\sim p \rightarrow q) \wedge (r \leftrightarrow s)$ are respectively

- (A) F, F
- (B) T, T
- (C) T, F
- (D) F, T

9. The expression $[(p \wedge \sim q) \vee q] \vee (\sim p \wedge q)$ is equivalent to

- (A) $p \vee q$
- (B) $p \wedge q$
- (C) $p \rightarrow q$
- (D) $p \leftrightarrow q$

10. Given p : A man is a judge, q : A man is honest

S_1 : If a man is a judge, then he is honest.

S_2 : If a man is a judge, then he is not honest.

S_3 : A man is not a judge or he is honest.

S_4 : A man is a judge and he is honest.

Then

- (A) $S_2 \equiv S_3$
- (B) $S_1 \equiv S_2$
- (C) $S_2 \equiv S_4$
- (D) $S_1 \equiv S_3$



TOGETHER WE LEARN



TOGETHER WE LEAD



TOGETHER WE SUCCEED

11. The statement pattern

$(p \wedge q) \wedge [(p \wedge q) \vee (\sim p \wedge q)]$ is equivalent to

- (A) q
- (B) $p \wedge q$
- (C) p
- (D) $p \vee q$

12. Let $a : \sim(p \wedge \sim r) \vee (\sim q \vee s)$ and

$b : (p \vee s) \leftrightarrow (q \wedge r)$.

If the truth values of p and q are true and r and s are false, then truth values of a and b are respectively

- (A) T, F
- (B) T, T
- (C) F, F
- (D) F, T

13.

S_1 : If -7 is an integer, then $\sqrt{-7}$ is a complex number.

S_2 : -7 is not an integer or $\sqrt{-7}$ is a complex number.

- (A) S_1 and S_2 are converse statements of each other
- (B) S_1 and S_2 are negations of each other
- (C) S_1 and S_2 are equivalent statements
- (D) S_1 and S_2 are contrapositive of each other

14. Negation of the statement: $3 + 6 > 8$ and $2 + 3 < 6$ is

- (A) $3 + 6 \leq 8$ or $2 + 3 < 6$
- (B) $3 + 6 < 8$ or $2 + 3 < 6$
- (C) $3 + 6 \leq 8$ or $2 + 3 \geq 6$
- (D) $3 + 6 > 8$ or $2 + 3 \geq 6$

15. “If two triangles are congruent, then their areas are equal.” is the given statement. The contrapositive of the inverse of the given statement is

- (A) If two triangles are not congruent, then their areas are equal
- (B) If two triangles are not congruent, then their areas are not equal
- (C) If areas of two triangles are equal, then they are congruent
- (D) If areas of two triangles are not equal, then they are congruent

16. The statement “If areas of two triangles are equal, then they are congruent” is

- (A) converse of the given statement
- (B) inverse of the given statement
- (C) contrapositive of the given statement
- (D) none of these

17. The negation of $p \wedge (q \rightarrow r)$ is

- (A) $\sim p \wedge (q \rightarrow r)$
- (B) $\sim p \vee (q \rightarrow r)$
- (C) $\sim p \vee (q \wedge \sim r)$
- (D) $p \vee (q \rightarrow \sim r)$



TOGETHER WE LEARN



TOGETHER WE LEAD



TOGETHER WE SUCCEED

18. If p : It is raining and q : It is pleasant, then the symbolic form of “It is neither raining nor pleasant” is

- (A) $\sim p \wedge \sim q$
- (B) $\sim p \vee \sim q$
- (C) $p \wedge q$
- (D) $p \vee q$

19. The negation of " $\forall x \in \mathbb{N}, x^2 + x$ is even number" is

- (A) $\forall x \in \mathbb{N}, x^2 + x$ is not an even number
- (B) $\forall x \in \mathbb{N}, x^2 + x$ is not an odd number
- (C) $\exists x \in \mathbb{N}$ such that $x^2 + x$ is an even number
- (D) $\exists x \in \mathbb{N}$ such that $x^2 + x$ is not an even number

20. If p: It is raining, q: It is pleasant. Then simplified form of the statement "It is not true, then weather is not pleasant" is

- (A) It is not raining or weather is pleasant
- (B) It is raining or weather is pleasant
- (C) It is raining or weather is not pleasant
- (D) It is raining and weather is pleasant

MHT-CET 2022 PYQs

1. The statement pattern $(p \rightarrow q) \vee (\sim p \rightarrow q)$ is logically equivalent to

- (A) $p \rightarrow q$
- (B) $p \rightarrow \sim q$
- (C) q
- (D) $p \vee q$

2. The negation of statement "I study or I fail" is

- (A) I study and I fail
- (B) I do not study and I do not fail
- (C) I study and I do not fail
- (D) I do not study and I fail

3. The statement pattern $[p \rightarrow (q \rightarrow r)] \leftrightarrow [(p \rightarrow q) \rightarrow r]$ is

- (A) equivalent to $p \leftrightarrow q$
- (B) a contingency
- (C) a contradiction
- (D) a tautology

4. The inverse of statement pattern $(p \vee q) \rightarrow (p \wedge q)$ is

- (A) $\sim(p \vee q) \rightarrow \sim(p \wedge q)$
- (B) $\sim(p \vee q) \rightarrow (p \wedge q)$
- (C) $(p \vee q) \rightarrow \sim(p \wedge q)$
- (D) $(\sim p \vee \sim q) \rightarrow (\sim p \wedge \sim q)$

5. If p and q each have truth values T and F respectively, then the truth value of the statement $p \vee q$ is

- (A) T
- (B) F
- (C) Both T and F
- (D) Cannot be determined

6. The statement pattern $\sim(p \leftrightarrow \sim q)$ is

- (A) equivalent to $(p \wedge q)$
- (B) a tautology
- (C) equivalent to $(\sim p) \wedge q$
- (D) a fallacy

7. The negation of the statement “The payment will be made if and only if the work is finished in time.” is

- (A) The work is finished in time and the payment is not made.
- (B) The payment is made and the work is not finished in time.
- (C) The work is finished in time and the payment is not made, or

the payment is made and the work is finished in time.

(D) Either the work is finished in time and the payment is not made, or the payment is made and the work is not finished in time.

8. If p and q are true statements and r and s are false statements, then the truth values of statement patterns $(p \rightarrow q) \vee r$ and $(p \vee s) \leftrightarrow (\sim r \wedge s)$ are respectively

- (A) T, F
- (B) T, T
- (C) F, F
- (D) F, T

9. The statement pattern $p \rightarrow \sim(p \wedge \sim q)$ is equivalent to

- (A) q
- (B) $\sim p \vee q$
- (C) $p \wedge q$
- (D) $p \vee q$

10. The logical statement

$(\sim(\sim p \vee q) \vee (p \wedge r)) \wedge (\sim q \wedge r)$ is equivalent to

- (A) $\sim p \vee r$
- (B) $(p \wedge \sim q) \vee r$
- (C) $(p \wedge r) \wedge \sim q$
- (D) $(\sim p \wedge \sim q) \wedge r$

11. The negation of statement “I study or I fail” is

- (A) I study and I fail
- (B) I do not study and I do not fail

- (C) I study and I do not fail
(D) I do not study and I fail

12. The statement pattern $p \rightarrow (q \rightarrow r)$ is

- (A) a contradiction
(B) a contingency
(C) equivalent to $p \rightarrow q$
(D) a tautology

13. The inverse of statement pattern $(p \vee q) \rightarrow (p \wedge q)$ is

- (A) $\sim(p \vee q) \rightarrow \sim(p \wedge q)$
(B) $\sim(p \vee q) \rightarrow (p \wedge q)$
(C) $(p \vee q) \rightarrow \sim(p \wedge q)$
(D) $(\sim p \vee \sim q) \rightarrow (\sim p \wedge \sim q)$

14. If p and q each have truth values F and T respectively, then the truth values of the statement patterns are respectively

- (A) T, F
(B) F, F
(C) F, T
(D) T, T

15. The statement pattern $\sim(p \leftrightarrow \sim q)$ is

- (A) equivalent to $(p \wedge q)$
(B) a tautology
(C) equivalent to $(\sim p) \wedge q$
(D) a fallacy

16. The negation of the statement “The payment will be made if and only if the work is finished in time.” is

- (A) The work is finished in time and the payment is not made.
(B) The payment is made and the work is not finished in time.
(C) The work is finished in time and the payment is not made, or the payment is made and the work is finished in time.
(D) Either the work is finished in time and the payment is not made, or the payment is made and the work is not finished in time.

17. Let p, q, r be three statements, then $[p \rightarrow (q \rightarrow r)] \leftrightarrow [(p \rightarrow q) \rightarrow r]$ is

- (A) equivalent to $p \leftrightarrow q$
(B) a contingency
(C) a tautology
(D) a contradiction

18. The logical statement $[\sim(\sim p \vee q) \vee (p \wedge r)] \wedge (\sim q \wedge r)$ is equivalent to

- (A) $(p \wedge \sim r) \vee q$
(B) $(p \wedge \sim q) \vee r$
(C) $\sim p \vee r$
(D) $p \wedge r$

19. The inverse of statement "If the surface area increases, then the pressure decreases" is

- (A) If the surface area does not increase, then the pressure does not decrease.
(B) If the pressure decreases, then the surface area increases.
(C) If the pressure does not decrease, then the surface area does not increase.

(D) If the surface area does not increase, then the pressure decreases.

20. Statement 1: If a quadrilateral is a square, then all of its sides are equal.

Statement 2: All the sides of a quadrilateral are equal, then it is a square.

- (A) Statement 2 is contrapositive of statement 1.
- (B) Statement 2 is negation of statement 1.
- (C) Statement 2 is inverse of statement 1.
- (D) Statement 2 is converse of statement 1.

21. If q is false and $p \wedge \sim q \leftrightarrow r$ is true, then _____ is a tautology.

- (A) $p \vee r$
- (B) $(p \wedge r) \rightarrow p \vee r$
- (C) $(p \vee r) \rightarrow p \wedge r$
- (D) $p \wedge r$

22. The statement pattern $p \rightarrow (q \rightarrow p)$ is

- (A) a contradiction
- (B) a contingency
- (C) a tautology
- (D) equivalent to $p \rightarrow q$

MHT-CET 2023 PYQs

1. The negation of statement “I study or I fail” is

- (A) I study and I fail
- (B) I do not study and I do not fail

- (C) I study and I do not fail
(D) I do not study and I fail

2. For three simple statements p , q and r , $p \rightarrow (q \rightarrow r)$ is logically equivalent to

- (A) $p \rightarrow (q \rightarrow r)$
(B) $\sim p \rightarrow (q \rightarrow r)$
(C) $p \rightarrow (q \wedge r)$
(D) $p \rightarrow (q \vee r)$

3. The negation of statement “The payment will be made if and only if the work is finished in time” is

- (A) The work is finished in time and the payment is not made.
(B) The payment is made and the work is not finished in time.
(C) The work is finished in time and the payment is not made, or the payment is made and the work is finished in time.
(D) Either the work is finished in time and the payment is not made, or the payment is made and the work is not finished in time.

4. Let p , q , r be three statements, then $[p \rightarrow (q \rightarrow r)] \leftrightarrow [(p \rightarrow q) \rightarrow r]$ is

- (A) equivalent to $p \leftrightarrow q$
(B) a contingency
(C) a tautology
(D) a contradiction

5. The logical statement

$[\sim(\sim p \vee q) \vee (p \wedge r)] \wedge (\sim q \wedge r)$ is equivalent to

- (A) $(p \wedge r) \wedge \sim q$
- (B) $p \wedge q$
- (C) $p \vee r$
- (D) $\sim p \vee r$

6. The given circuit is equivalent to

- (A) S_3
- (B) S_2
- (C) S_1
- (D) S_1'

7. The inverse of the statement “If the surface area increases, then the pressure decreases” is

- (A) If the surface area does not increase, then the pressure does not decrease.
- (B) If the pressure decreases, then the surface area increases.
- (C) If the pressure does not decrease, then the surface area does not increase.
- (D) If the surface area does not increase, then the pressure decreases

8. If p and q are true statements and r and s are false statements, then the truth values of statement patterns $(p \rightarrow q) \vee r$ and $(p \vee s) \leftrightarrow (\sim r \wedge s)$ are respectively

- (A) T, F
- (B) T, T
- (C) F, F
- (D) F, T

9. The statement pattern $p \rightarrow \sim(p \wedge \sim q)$ is equivalent to

- (A) q
- (B) $\sim p \vee q$
- (C) $p \wedge q$
- (D) $p \vee q$

10. The logical statement

$(\sim(\sim p \vee q) \vee (p \wedge r)) \wedge (\sim q \wedge r)$ is equivalent to

- (A) $\sim p \vee r$
- (B) $(p \wedge \sim q) \vee r$
- (C) $(p \wedge r) \wedge \sim q$
- (D) $(\sim p \wedge \sim q) \wedge r$

11. If truth value of logical statement $(p \leftrightarrow \sim q) \rightarrow (\sim p \wedge q)$ is false, then the truth values of p and q are respectively

- (A) F, T
- (B) T, T
- (C) T, F
- (D) F, F

12. The contrapositive of “If x and y are integers such that xy is odd, then both x and y are odd” is

- (A) If both x and y are odd integers, then xy is odd.
- (B) If both x and y are even integers, then xy is even.
- (C) If x or y is an odd integer, then xy is odd.
- (D) If both x and y are not odd integers, then xy is not odd.

13. The inverse of the statement “If the surface area increases, then the pressure decreases” is

- (A) If the surface area does not increase, then the pressure does not decrease.
- (B) If the pressure decreases, then the surface area increases.
- (C) If the pressure does not decrease, then the surface area does not increase.
- (D) If the surface area does not increase, then the pressure decreases.

14. Statement 1: If a quadrilateral is a square, then all of its sides are equal.

Statement 2: All the sides of a quadrilateral are equal, then it is a square.

- (A) Statement 2 is contrapositive of statement 1.
- (B) Statement 2 is negation of statement 1.
- (C) Statement 2 is inverse of statement 1.
- (D) Statement 2 is converse of statement 1.

15. The given following circuit is equivalent to

- (A) S_1'
- (B) S_2'
- (C) S_1
- (D) S_2



TOGETHER WE LEARN



TOGETHER WE LEAD



TOGETHER WE SUCCEED

16. The expression $(p \wedge \sim q) \vee q \vee (\sim p \wedge q)$ is equivalent to

- (A) $\sim p \vee q$
- (B) $p \wedge q$
- (C) $p \vee q$
- (D) $p \vee \sim q$

17. Negation of inverse of the following statement pattern $(p \wedge q) \rightarrow (p \vee \sim q)$ is

- (A) p
- (B) $\sim q$
- (C) $\sim p$
- (D) q

18. If q is false and $p \wedge \sim q \leftrightarrow r$ is true, then _____ is a tautology.

- (A) $p \vee r$
- (B) $(p \wedge r) \rightarrow p \vee r$
- (C) $(p \vee r) \rightarrow p \wedge r$
- (D) $p \wedge r$

19. Negation of contrapositive of statement pattern $(p \vee \sim q) \rightarrow (p \wedge \sim q)$ is

- (A) $(\sim p \wedge q) \vee (p \wedge \sim q)$
- (B) $(\sim p \vee q) \wedge (p \vee \sim q)$
- (C) $(p \wedge \sim q) \vee (p \vee \sim q)$
- (D) $(\sim p \vee \sim q) \wedge (p \vee q)$

20. The statement $[(p \rightarrow q) \wedge \sim q] \rightarrow r$ is tautology, when r is equivalent to

- (A) $p \wedge \sim q$
- (B) $q \vee p$
- (C) $p \wedge q$
- (D) $\sim q$

21. The negation of the statement “The number is an odd number if and only if it is divisible by 3” is

- (A) The number is an odd number but not divisible by 3 or the number is divisible by 3 but not odd.
- (B) The number is not an odd number if it is not divisible by 3.
- (C) The number is not an odd number but it is divisible by 3.
- (D) The number is not an odd number or is not divisible by 3.

22. If the statement $p \leftrightarrow (q \rightarrow p)$ is false, then true statement/statement pattern is

- (A) p
- (B) $p \rightarrow (p \vee \sim q)$
- (C) $p \wedge (\sim p \wedge q)$
- (D) $(p \vee \sim q) \rightarrow p$

23. The statement $[p \wedge (q \vee r)] \vee [\sim r \wedge \sim q \wedge p]$ is equivalent to

- (A) $\sim r$
- (B) p
- (C) $\sim q$
- (D) q

TEAM
ALPHA



TOGETHER WE LEARN



TOGETHER WE LEAD



TOGETHER WE SUCCEED

MHT-CET 2024 PYQs

1. If the statement $p \vee \sim(q \wedge r)$ is false, then the truth values of p , q and r are respectively

- (A) F, T, F
- (B) T, F, F

- (C) F, T, T
(D) F, F, T

2. If Statement I: If the work is not finished on time, the contractor is in trouble.

If Statement II: Either the work is finished on time or the contractor is in trouble, then

- (A) Statement II is negation of statement I.
(B) Statement II is converse of statement I.
(C) Statement II and statement I are equivalent.
(D) Statement II is inverse of statement I.

3. The negation of contrapositive of the statement $p \rightarrow (\sim q \wedge r)$ is

- (A) $(\sim q \vee \sim r) \wedge \sim p$
(B) $(q \vee \sim r) \wedge p$
(C) $(q \wedge \sim r) \vee p$
(D) $(\sim q \wedge \sim r) \vee \sim p$

4. Which one of the following is the pair of equivalent circuits?

(i), (ii), (iii), (iv), (v) are the given circuits.

- (A) (i) and (ii)
(B) (ii) and (iv)
(C) (iii) and (v)
(D) (i) and (iii)

5. Consider the following statements:

p: The switch S_1 is closed

q: The switch S_2 is closed

r: The switch S_3 is closed

Then the switching circuit represented by the statement

$(p \wedge q) \vee (\sim p \wedge (\sim q \vee p \vee r))$ is

(A) [Circuit A]

(B) [Circuit B]

(C) [Circuit C]

(D) [Circuit D]

6. If p and q are statements, then _____ is a contingency.

(A) $p \wedge \sim p$

(B) $p \vee \sim p$

(C) $p \vee q$

(D) $(p \wedge (p \rightarrow q)) \rightarrow q$

7. The new switching circuit for the following circuit by simplifying the given circuit is

(A) S_2

(B) S_3

(C) S_1

(D) S_1'

8. If $(p \wedge \sim q) \wedge (p \wedge r) \rightarrow \sim p \vee q$ is false, then the truth values of p, q and r are respectively

(A) T, T, T

(B) F, F, F

(C) T, F, T

(D) F, T, F

ANSWER KEY

MHT-CET 2021

1.C	2.C	3.D	4.C	5.D	6.A	7.A
8.B	9.A	10.D	11.B	12.C	13.C	14.C
15.C	16.D	17.B	18.C	19.D	20.D	

ANSWER KEY

MHT-CET 2022

1.C	2.B	3.D	4.A	5.A	6.C	7.D
8.D	9.D	10.B	11.B	12.D	13.A	14.D
15.C	16.A	17.C	18.D	19.A	20.A	21.D
22.C						

ANSWER KEY

MHT-CET 2023

1.B	2.B	3.D	4.C	5.A	6.D	7.A
8.D	9.B	10.C	11.C	12.D	13.A	14.D
15.A	16.C	17.B	18.B	19.B	20.D	21.A
22.B	23.B					

ANSWER KEY

MHT-CET 2024

1.C	2.C	3.B	4.D	5.C	6.C	7.C
-----	-----	-----	-----	-----	-----	-----

8.C

